

Small sequence changes can close near-term sensitivity gaps

■ measured requirement ➡ minimum requirement ■ one-mutation gain observed

Ciprofloxacin

Clinical drug monitoring

EC50 22.4 μ M vs plasma Cmax $\sim$ 9 μ M

L-carnitine

Metabolic health

no plateau by 200 μ M; plasma to 75 μ M

Ergothioneine

Nutrition / oxidative stress

no plateau by 200 μ M; RBC to 160 μ M

Thiamine

Vitamin status

no plateau by 200 μ M; blood to 0.195 μ M

DEHP

Environmental exposure

EC50 34.5 μ M; background serum $<$ 0.005 μ M

